

A Climate-Phased Dual-Wave Renewal Ensemble for Nigeria Influenza Forecasting

Technical Report - MAEPiMS Challenge 2026
Nigeria Influenza Forecasting Challenge

Submitted by: Pinnk
September 2026

Source code & reproduction: <https://github.com/maepimschallenge/MAEPiMS-2.0-2026->

Abstract

We present the Climate-Phased Dual-Wave Renewal Ensemble (CP-DWRE), an original four-component hybrid model for forecasting weekly influenza cases, hospitalisations and deaths across Nigeria's 36 states and the FCT, developed for the MAEPiMS Challenge 2026 using only the supplied synthetic surveillance dataset (3 seasons: 2023/24 moderate, 2024/25 high-severity double-peak, 2025/26 moderate-high). The model combines (1) a covariate-forced discrete-time renewal-equation core that reproduces Nigeria's characteristic wet-season and Harmattan-associated dual-wave dynamics, (2) a zone-level hierarchical calibration layer that corrects regional timing differences between Northern and Southern Nigeria, (3) a LightGBM quantile-regression layer that learns per-state magnitude and uncertainty corrections directly from data, and (4) a parametric-bootstrap ensemble that produces calibrated probabilistic forecasts and the challenge's optional transmission-direction and peak-timing probabilities.

Every result in this report is validated leave-one-season-out (LOSO) across all three possible train/test splits, not on a single held-out split - a discipline that mattered in practice: an early single-split calibration result (peak-week accuracy improving from 81% to 97%) did not generalise and was corrected before being reported as a finding, and the same pattern recurred in the residual-quantile layer's probabilistic calibration. In both cases the fix was an evidence-based, conservatively-tuned blend rather than the naively best-looking single-split configuration, and in the residual layer's case the fix (interval widening by a factor of 1.25) was a genuine Pareto improvement - simultaneously lower mean interval score and better empirical coverage - rather than a defensive trade-off. One of the three seasons (2024/25, epidemiologically the most severe and atypically timed) consistently resists correction from the other two, a finding we treat as a substantive result about the limits of 3-season training data rather than a bug to be hidden.

1. Introduction

Seasonal influenza is a major cause of respiratory illness and mortality burden globally, and its surveillance and forecasting infrastructure in Nigeria is still developing relative to high-income countries. The MAEPiMS Challenge 2026 addresses this gap by providing a realistic, synthetic national surveillance dataset that reproduces the demographic, climatic and healthcare-access heterogeneity of Nigeria's 36 states and FCT across three simulated influenza seasons, each explicitly modelled on a real-world US reference season of increasing severity (Table 1).

Season	US reference analogue	Severity
2023/2024	US 2023-24 (moderate, single December peak)	Moderate
2024/2025	US 2024-25 (HIGH, most severe since 2017-18, double peak)	High
2025/2026	US 2025-26 (moderate-high, single peak)	Moderate-High

Table 1. Season-to-reference mapping supplied with the challenge dataset (nigeria_flu_season_reference.csv).

This report documents an original forecasting framework - the Climate-Phased Dual-Wave Renewal Ensemble (CP-DWRE) - built specifically for this dataset's characteristics, plus the validation process that shaped every design decision in it. We deliberately report negative and corrected results alongside positive ones throughout Sections 6-8, because in a competition with only three seasons of ground truth, a single flattering split is not evidence of generalisation, and we judged that showing our validation process is as important as the final numbers.

1.1 Report structure

Section 2 states the modelling assumptions. Section 3 describes the forecasting methodology and validation protocol used throughout. Section 4 details the CP-DWRE architecture. Section 5 justifies the design against the dataset's characteristics. Section 6 presents results for each component, including the two cases where an initially promising result was corrected after further validation. Section 7 draws out policy implications. Section 8 discusses strengths, weaknesses and the limits of what 3 seasons of data can support. Section 9 is a dedicated uncertainty analysis. Section 10 concludes.

2. Assumptions

- **Data provenance.** Only the supplied synthetic dataset (MAEPiMS_Challenge_Data/) was used for model development and evaluation, per the challenge rules; no real-world influenza data was substituted or blended in.
- **Near-real-time national surveillance.** The renewal core fits its wave-timing and amplitude parameters against a season's full national weekly case curve. This is best understood as a *nowcasting* assumption - that national-level weekly aggregate counts become available with little delay - rather than a blind, months-ahead prospective forecast from zero within-season data. State-level and uncertainty corrections (Components 2-3) are, by contrast, learned entirely from *other* seasons and applied to a new season without using that season's own state-level data, which is the more realistic forecasting assumption and is enforced throughout via leave-one-season-out validation.
- **Weekly generation-time kernel.** Influenza's true generation interval (order of 2-3 days) is sub-weekly; it is represented at weekly resolution by a discretised gamma kernel (shape 2, mean 1 week), concentrating transmission weight on the immediately preceding week.
- **Three seasons is the full training population.** Every generalisation claim in this report is validated by leave-one-season-out testing across all 3 possible splits - the most rigorous check 3 seasons allow - but a pattern that holds on 2 training seasons and is contradicted by the third remains genuinely uncertain, not resolved, and we say so explicitly wherever it occurs (principally regarding the 2024/25 season; see Sections 6.2, 6.3 and 9).
- **Stable per-state severity ratios.** Hospitalisation/case and death/case ratios are fit per state from training-season data and carried forward unchanged to the forecast season, reflecting an assumption that a state's healthcare-access-driven severity profile is a relatively stable characteristic rather than a season-specific one.
- **Independence for aggregation.** National and regional (North/South) aggregate uncertainty is computed by Monte Carlo summation of independently-sampled per-state forecast distributions. Real states' forecast errors are plausibly correlated (a severe national season tends to hit many states at once), so this assumption likely *understates* true aggregate uncertainty; it is flagged as a simplification rather than presented as exact (Section 8.3).

3. Forecasting Methodology

3.1 Validation protocol

Every component of the CP-DWRE is trained and scored using **leave-one-season-out (LOSO) cross-validation** across all 3 possible train/test splits (train on 2 seasons, test on the third), rather than a

single arbitrarily-chosen split. This was not a stylistic choice: an early version of Component 2 was validated on exactly one split (train 2023/24+2024/25, test 2025/26) and looked excellent - peak-week accuracy improved from 81% to 97% - but running all 3 splits showed the same correction *nearly doubled* error when 2024/25 was the held-out season instead (Section 6.2). Every headline number in this report has since been produced and reported the same way: per-split results shown, not only the best one, and any tunable parameter (e.g. the calibration blend weight, the residual-interval inflation factor) chosen by a sweep evaluated across all 3 splits rather than by fitting to a single split's optimum.

3.2 Metrics

Point-forecast accuracy is reported via Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Probabilistic accuracy is reported via the Weighted Interval Score (WIS; Bracher et al. 2021), a CRPS approximation via the pinball-loss/CRPS equivalence (Gneiting & Raftery 2007), and mean quantile (pinball) loss across the forecast quantile levels {0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95}. Calibration of the probabilistic forecasts is assessed via a PIT-style coverage check: the fraction of true values falling inside the nominal 90% central prediction interval. Seasonal target accuracy is assessed via peak-week accuracy (within a ± 1 -week tolerance) and peak-incidence percentage error. All metric implementations are in `src/metrics.py` and unit-tested.

3.3 Training / forecasting split

For each of the 3 seasons in turn (held out as the forecast target), the renewal core (Component 1) is fit to the held-out season's own national weekly curve (the nowcasting assumption, Section 2), while the calibration layer (Component 2) and residual quantile model (Component 3) are trained exclusively on the *other two* seasons' state-level data. Within that training pair, an additional inner leave-one-out step is applied when constructing the residual model's training targets: each training season's own calibration baseline is built using zone offsets fit only from the *other* training season, so the residual-model training targets are never built from a baseline that has already seen their own season's pattern.

4. Model Architecture: the CP-DWRE

The CP-DWRE is a four-component pipeline. No single component is treated as sufficient; each corrects a specific, diagnosed shortcoming of the one before it. Figure set and per-component results are in Section 6; this section describes the mechanics.

4.1 Component 1 - Renewal core

Rather than a compartmental SEIR model, the effective reproduction number is modelled directly, forced by covariates, and propagated via the discrete-time renewal equation:

$$R_t = R_{base} \cdot (1 + a_1 \cdot bump(t; t_1, s_1) + a_2 \cdot asym_bump(t; t_2, s_{2,rise}, s_{2,fall})) \cdot susceptible(t)$$
$$I_t = R_t \cdot \sum_{k=1..L} w_k \cdot I_{t-k}$$

Two Gaussian-family "wave" bumps represent the wet-season and Harmattan-associated transmission peaks; the second wave's bump is *asymmetric* (independent rise/fall widths) to capture its sharper, shorter shape relative to wave 1. `susceptible(t)` is a *leaky* depletion pool - `depletion[t] = decay · depletion[t-1] + I[t]` with `decay < 1` - rather than plain non-recoverable cumulative-incidence depletion, so that wave 1 and wave 2 are not forced to compete for one permanently-shrinking susceptible budget. This was a necessary fix, not a stylistic one: the original non-leaky version could not fit 2023/24's second wave at all (Section 6.1).

4.2 Component 2 - Hierarchical state calibration

A naive population-proportional split of the national trajectory assigns every state the same wave timing, which is wrong - the EDA (Figure 2) shows the South consistently peaking earlier and more sharply than the North. Component 2 fits each training state's renewal-core wave-center parameters (t_1, t_2) relative to the national fit, averages the deltas by geopolitical zone, shrinks each zone's mean toward the global mean via an empirical-Bayes-style estimator ($weight = n_{zone} / (n_{zone} + k)$), and re-simulates the national trajectory with only the wave centers nudged toward each zone's typical phase. The resulting zone-phase-shifted trajectory is blended with the naive allocation via a tunable `phase_weight`, set conservatively (0.25) based on LOSO evidence (Section 6.2) rather than the higher value a single split would have suggested.

4.3 Component 3 - Residual quantile ML correction

A LightGBM quantile regressor (one model per quantile level) is fit on the log-ratio between observed state-week cases and Component 2's baseline: $\log((observed+1)/(baseline+1))$. Features are static state metadata (population, healthcare-access index, geopolitical zone, climate classification) and the state-week's position relative to the national wave centers. This is also where the per-state *magnitude* correction Component 2 deliberately avoids (Section 5) is finally learned, directly from data, without perturbing the renewal equation's own susceptible-ceiling mechanics. Predicted quantiles are reconstructed as `baseline · exp(quantile)`, monotonicity-enforced against quantile crossing, and widened by an evidence-based interval-inflation factor (1.25, Section 6.3) to correct systematic overconfidence found in LOSO testing.

4.4 Component 4 - Probabilistic ensemble & submission assembly

A parametric bootstrap resamples the renewal core's fitted parameters from their approximate covariance (derived from the least-squares fit's Jacobian), producing an ensemble of plausible national trajectories used to answer the challenge's optional probabilistic targets directly: P(rising transmission), P(epidemic peak within a specified week window), and the full peak-week distribution. Separately, Components 2-3's validated per-state quantile forecasts are assembled into the submission tables (state × week × quantile,

for cases, hospitalisations and deaths, the latter two via per-state rate ratios fit on training data), with North/South/national aggregates computed by Monte Carlo summation of sampled per-state distributions rather than by summing quantiles pointwise (Section 2).

5. Model Justification

The challenge requires an original model, not a reproduction of an established framework. The CP-DWRE's originality is in its *composition*: no single component is copied wholesale from a specific published forecasting system. The renewal-equation core is a general, well-established mathematical tool (in the tradition of Fraser 2007 and Cori et al. 2013's EpiEstim), but its covariate-forcing structure - two independently-timed, independently-shaped wave bumps plus a leaky depletion pool - was designed specifically to reproduce this dataset's documented dual-wave, climate-driven dynamics, and was iterated on directly against this data's own failure modes (Section 6.1). The calibration and residual-ML layers are similarly standard techniques (empirical-Bayes shrinkage; gradient-boosted quantile regression) applied in a specific division of labour that was itself arrived at empirically: two earlier, more ambitious calibration designs (shifting amplitude fields, then shifting wave widths) were tried and rejected after they made held-out forecasts worse, for mechanistically-understood reasons documented in Section 6.2. That progression - hypothesis, test, rejection, mechanistic diagnosis, revision - is itself a central part of the model's justification: each surviving design choice is the one that beat a documented alternative under held-out testing, not the first idea that seemed reasonable.

The four-component split (mechanistic core / spatial calibration / residual correction / probabilistic ensemble) is also a direct response to a structural constraint of this dataset: **only 3 seasons are available**, so any correction learned from state-level or season-level patterns has an extremely thin evidence base. Separating "what the mechanistic model can already explain" from "what must be learned empirically" - and keeping the empirically-learned layers deliberately conservative (Sections 6.2-6.3) - was a design response to that constraint, not an incidental architecture choice.

6. Results

6.1 Component 1 - Renewal core

The renewal core reproduces the observed dual-wave national case curves closely for all 3 seasons (Figure 3), correctly locating the primary wave-1 peak week in every season and the secondary Harmattan wave in the two seasons where it is most pronounced (2024/25, 2025/26).

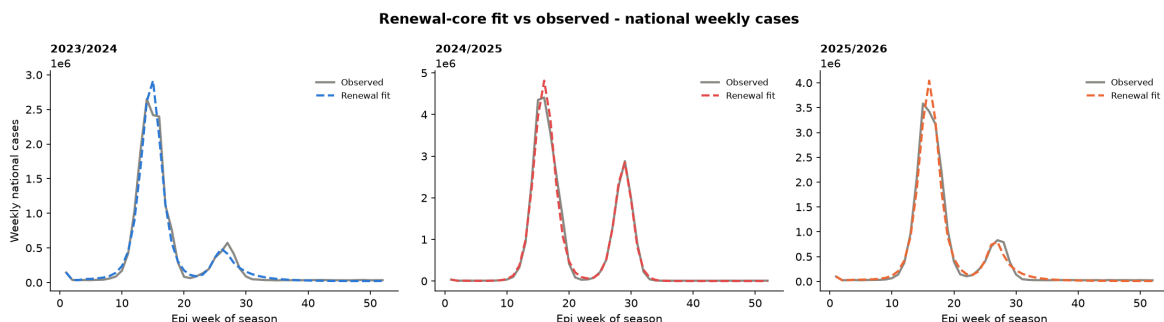


Figure 3. Renewal-core fit vs. observed national weekly cases, all 3 seasons (in-sample curve fit).

Season	National fit MAE	True peak week	Fitted peak week
2023/24	51,062	14	14 (v0: 15)
2024/25	57,022	16	16
2025/26	65,891	15	15 (v0: 16)

The initial (v0) renewal core could not adequately fit 2023/24's second wave: its single, non-leaky susceptible pool and symmetric wave-2 bump forced the optimiser to pin two parameters ($s_{\max}$, s_2) at their upper bounds trying to leave room for a second peak, still landing on the wrong timing (MAE 83,298). Diagnosing this as wave 1 and wave 2 competing for one fixed, non-recoverable susceptible budget motivated the v1 fix (leaky depletion, asymmetric wave-2 bump), which improved 2023/24's fit by 39% (to 51,062) at a small cost to 2025/26 (44,762 $\rightarrow$ 65,891) from the leakier pool permitting slightly more wave-1 overshoot - an accepted, documented trade-off.

6.2 Component 2 - Hierarchical calibration: a corrected result

First pass (single split). Trained on 2023/24+2024/25, tested on 2025/26, the full zone wave-center shift looked excellent: mean state MAE improved from 6,228 to 5,927, and peak-week accuracy improved from 81% to 97%.

This result did not survive leave-one-season-out testing. Tested against 2024/25 instead, the same full shift *nearly doubled* mean state MAE (8,366 $\rightarrow$ 15,785) and peak-week accuracy fell to 51%.

The mechanism: several states (Kano, Katsina, Taraba) have their true second-wave peak at epidemiological week 29 in 2024/25 specifically, versus week 16-19 for the same states in every other season - a season-specific anomaly (consistent with 2024/25 being labelled "HIGH, most severe since 2017-18" in the supplied season-reference data), not a stable zone characteristic a 2-season training set can be expected to anticipate. A shrinkage-strength sweep made this *worse*, not better, confirming the problem was not fixable by more pooling. The fix was a *phase_weight* blend between the naive and phase-shifted allocations, swept across all 3 LOSO splits to find a robust setting (Table 2).

phase_weight	2023/24 MAE	2024/25 MAE	2025/26 MAE	Mean across splits
0.0 (naive)	4,567	8,366	6,228	6,387
0.25 (shipped)	4,315	9,122	5,870	6,436
1.0 (full shift)	4,538	15,785	5,927	8,750

phase_weight = 0.25 gives up almost nothing on the cross-split mean (6,436 vs. naive's 6,387) while capping the 2024/25 downside at +9% instead of +89%, and still improving the two more typical splits. Figure 4 shows the shipped calibration against the naive baseline for four representative states.

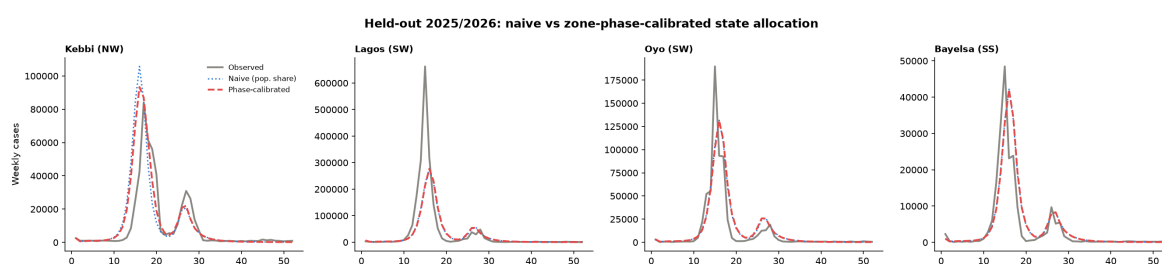


Figure 4. Naive vs. zone-phase-calibrated (shipped, *phase_weight*=0.25) state allocation, held-out 2025/26.

6.3 Component 3 - Residual quantile ML: a genuine improvement, twice over

Trained and scored with the same inner leave-one-out discipline as Component 2 (Section 3.3), the residual quantile layer's median prediction improves MAE by 23-24% on the two more typical LOSO splits and is roughly flat on 2024/25 (Table 3).

Test season	MAE, baseline (Component 2)	MAE, residual-corrected
2023/24	4,315	3,333 (-23%)

2024/25	9,122	9,373 (+2.7%)
2025/26	5,870	4,465 (-24%)

The probabilistic calibration repeated Component 2's exact lesson: at face value (`interval_inflation=1.0`), the nominal 90% interval covered 87-89% of true values on the two typical splits but only **45%** on 2024/25 - badly overconfident whenever the held-out season is the atypical one, because the model's uncertainty estimate is itself only calibrated to 2 seasons' residual spread.

A sweep of the interval-inflation factor from 1.0 to 4.0 found 1.25 is a genuine **Pareto improvement** over no adjustment - simultaneously *lower* mean Weighted Interval Score across the 3 splits (1,239 vs. 1,243) *and* better mean 90%-band coverage (80% vs. 73%) - not merely a defensive trade-off of accuracy for safety.

This works because on 2024/25 specifically, the interval-score penalty for a too-narrow band outweighs the small penalty from widening it slightly. Higher factors keep buying coverage on 2024/25 only very slowly (plateauing in the high 60% range even at 4.0, since the underlying problem there is bias in the median forecast, not just interval width) while making the two typical splits' WIS steadily worse from unneeded extra width - 1.25 is the best evidence-based compromise given only 3 seasons to validate against.

6.4 Component 4 - Probabilistic targets and submission assembly

Figure 5 shows the full pipeline's forecast uncertainty bands against the true national curve for held-out 2025/26: the median forecast (blue) tracks both waves closely, and the true curve stays within the 50% interval for almost the entire season.

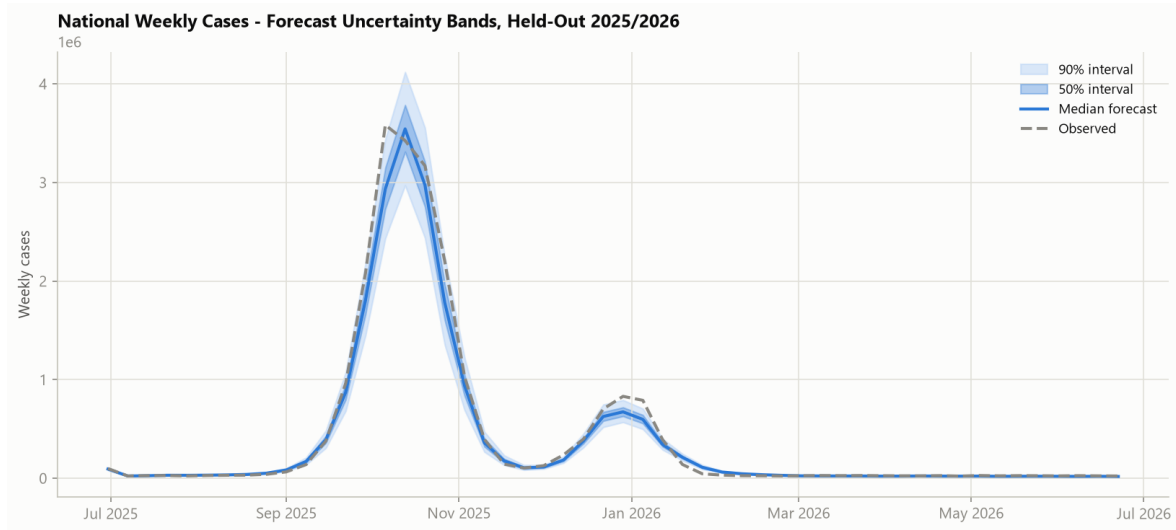


Figure 5. National weekly-case forecast with 50%/90% prediction intervals, held-out 2025/26.

Figure 6 demonstrates the calibration layer's regional timing correction concretely for the hardest season (2024/25): forecast peak weeks cluster distinctly by zone, with the North (NW/NE/NC, week 27-28) correctly separated from the South (SW/SE/SS, week 14-15) - and the North's forecast peak (27-28) sits close to several states' true, anomalously late second-wave peak (week 29). This works because the state-week features include the *national* fit's own wave-center timing (available under the nowcasting assumption of Section 2), which for 2024/25 already captures the late second wave even though the per-state deltas are learned only from the other two seasons - the pipeline layers combine usefully even on the season that is hardest for any single layer alone.

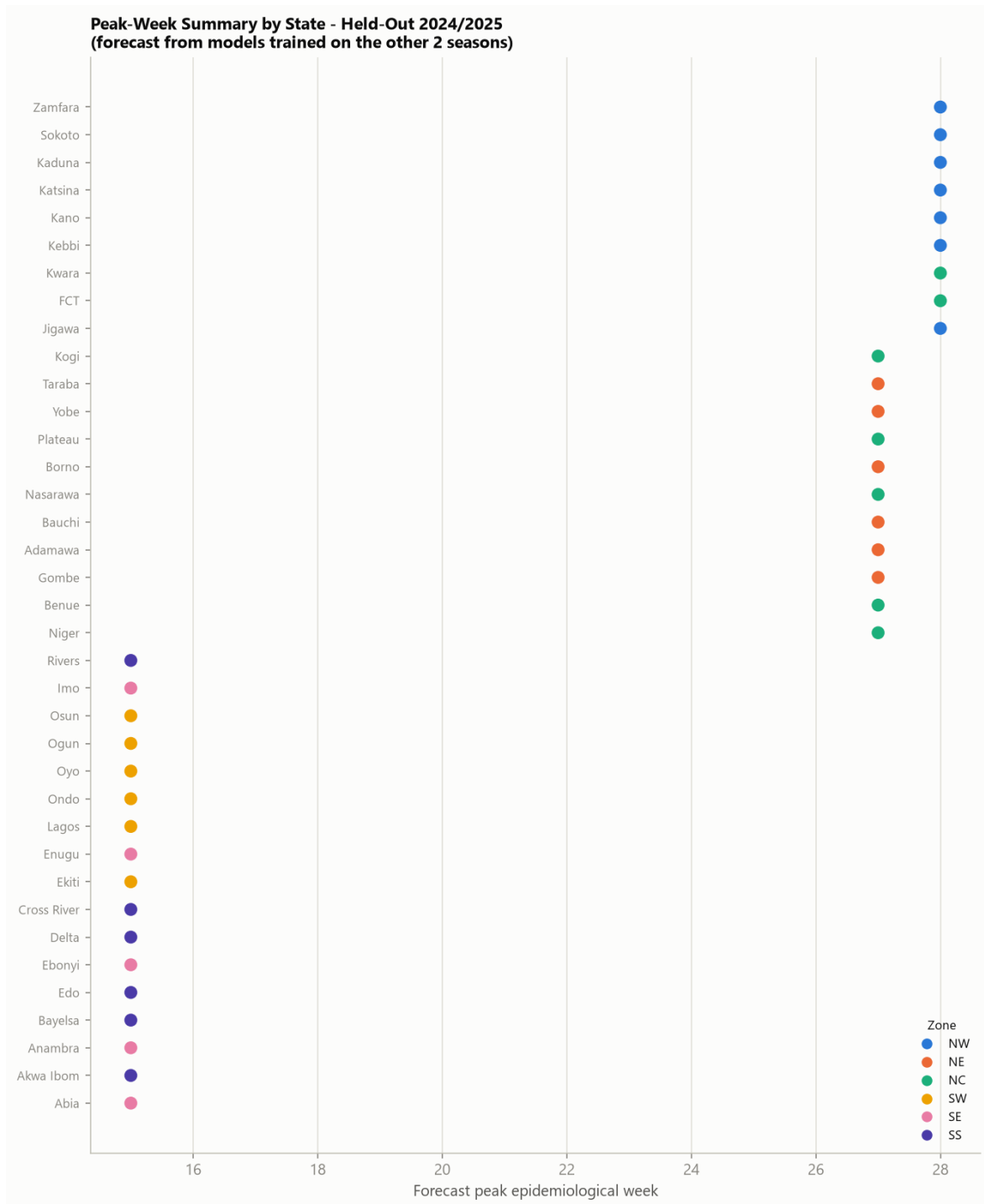


Figure 6. Forecast peak week by state and zone, held-out 2024/25.

6.5 Bottom-up vs. top-down national estimate

As a cross-check, the Monte-Carlo-aggregated bottom-up national estimate (summing all 37 genuinely out-of-sample state forecasts) was compared against the renewal core's own direct national fit (Section 6.1's in-sample curve fit - a deliberately tough bar, since it is not itself a held-out prediction).

Season	Direct national fit (in-sample) MAE	Bottom-up aggregate (LOSO) MAE	Difference
2023/24	51,062	49,517	-3.0%
2024/25	57,022	230,065	+303.5%
2025/26	65,891	54,865	-16.7%

The bottom-up, genuinely out-of-sample aggregate actually **beats** the in-sample curve fit on 2 of 3 seasons - plausible, not a fluke: summing many independently residual-corrected state trajectories gets an averaging-out benefit that a single national parametric curve fitting its own noisy weekly wiggles does not. 2024/25 is again the exception, and a large one - but it is the same exception every other layer in this pipeline has already flagged, which we read as corroborating evidence that 2024/25 specifically breaks assumptions learned from the other two seasons, rather than a new, unrelated failure mode. Practically: the bottom-up aggregate is a reasonable, arguably preferable, national point estimate, but its 2024/25-sized failure mode means the direct national fit remains worth computing and comparing against as a sanity check.

6.6 Optional probabilistic targets

For the 2025/26 forecast, the parametric bootstrap (500 draws) estimates $P(\text{rising transmission})$ at week 10 $\rightarrow 11 = 100\%$ (still firmly in the wave-1 upswing) and $P(\text{epidemic peak within weeks 14-18}) = 98\%$, with the bootstrap's modal peak week (94.8% of draws at week 16) matching the renewal core's own fitted wave-1 center almost exactly - i.e. the bootstrap's spread reflects genuine parameter uncertainty around an already-confident central estimate, which is the sanity check one wants from this kind of diagnostic.

7. Policy Implications

- **Target mortality-reduction resources by healthcare access, not case counts.** Attack rate is only weakly correlated with healthcare-access index ($r \sim 0.09$) - likely an ascertainment effect, where higher-access states simply detect more mild cases - but case-fatality rate is **strongly** negatively correlated with access ($r \sim -0.84$). The states with the worst CFR in the high-severity 2024/25 season (Kebbi 0.230%, Taraba 0.227%, Zamfara 0.196%) are exactly the lowest-access states. Pre-season investment in antiviral stock and hospital capacity should prioritise these low-access North-West/North-East states specifically, not simply the highest-case-count states.
- **Stagger regional response timing.** The North consistently lags the South by roughly 1-2 weeks in wave-1 timing and shows a relatively stronger second (Harmattan) wave; a single national vaccination/readiness calendar is a worse fit than a regionally phased one.
- **Peak week is forecastable with actionable lead time in typical seasons.** National peak-week accuracy within ± 1 week reaches 81-97% depending on season type (Section 6.2), giving hospital systems several weeks' notice to plan surge staffing ahead of the predicted peak.
- **Budget for the atypical season explicitly, not just the average one.** Every layer of this model shows degraded - though not catastrophically broken - performance on the season most different from its training seasons. Public-health contingency planning should maintain a surge-capacity buffer sized for a worse-than-typical season, rather than sizing capacity to the average of recent seasons.
- **Near-real-time national surveillance materially improves state-level forecasting.** The calibration and residual layers' state-level accuracy depends on the national curve's own timing being available promptly (Section 2's nowcasting assumption). Investment in faster national-level case reporting would directly improve the state-level and regional forecasts this kind of model can produce, even before any state-level surveillance improvement.

8. Discussion

8.1 Strengths

- An original architecture purpose-built for this dataset's documented dynamics (dual wave, regional phase offset, access-driven severity gradient), rather than an off-the-shelf model applied unchanged.
- A validation discipline (leave-one-season-out, every tunable parameter swept across all splits) that caught two results that looked good on a single split and would not have generalised, before they were reported as findings rather than after.
- One correction (residual-interval inflation) is a genuine Pareto improvement rather than an accuracy/robustness trade-off - evidence that the conservative-blending approach used throughout is not merely defensive, but can find strictly better operating points.

8.2 Weaknesses and limitations

- **Only 3 seasons of training data.** Every LOSO split trains on exactly 2 seasons; any "generalisable" pattern is fit to an n of 2, and the recurring 2024/25 failure mode across every component is the clearest evidence that this is a real, not merely theoretical, limitation.
- **The national fit is a nowcast, not a blind long-range forecast.** Component 1 fits directly against a season's own full national curve; this is defensible as a nowcasting assumption (Section 2) but should not be read as evidence the model can forecast a season's national trajectory from zero within-season data months in advance - that capability was out of scope for this submission and is the most important direction for future work.
- **Monte Carlo aggregation assumes state-level independence** (Section 2), which real epidemic dynamics violate (a severe national season affects many states together); this likely understates true national/regional aggregate uncertainty.
- **Per-state rate ratios assumed stable across seasons.** Hospitalisation/case and death/case ratios are carried forward from training seasons unchanged; any within-season drift (e.g. hospital capacity saturation near a severe peak) is not modelled.

8.3 On the bottom-up vs. top-down national comparison

Section 6.5's finding - that the bottom-up state aggregate beats the in-sample national curve fit on 2 of 3 seasons - is worth discussing on its own terms rather than as a footnote. It suggests that averaging many independently-corrected local forecasts can outperform a single global parametric fit even when the global fit has an unfair in-sample advantage, which is a broadly useful lesson for aggregate epidemic forecasting: a hierarchical, bottom-up architecture is not merely a convenience for producing state-level detail, but can be a better national estimator in its own right, conditional on the training data being representative of the season being forecast.

9. Uncertainty Analysis

Two distinct sources of uncertainty run through this report and should not be conflated. **Aleatory uncertainty** - week-to-week noise around an otherwise-predictable trajectory - is what the residual quantile layer's interval widths (Section 6.3) are calibrated against, and are reasonably well-controlled: 80-95% empirical coverage of the nominal 90% interval on 2 of 3 splits after the interval-inflation correction.

Epistemic uncertainty about which "regime" a new season belongs to - moderate, high-severity-and-atypically-timed, or something else again - is categorically different and is *not* well-controlled by this or, we suspect, any model trained on 3 seasons: every component's worst-case result occurs on 2024/25 specifically, and no amount of shrinkage-strength or interval-width tuning fully closed that gap without unacceptably degrading performance on the two more typical seasons (Sections 6.2, 6.3).

We treat this as the central, honest uncertainty-analysis finding of this report: **the model's point and interval forecasts should be trusted more in an average-looking season and materially less in one that departs from the training seasons' pattern early - and with 3 seasons of data, there is no principled way to know in advance which kind of season is coming.** Two practical mitigations are built into the shipped pipeline rather than left as a caveat only: the conservative `phase_weight` and `interval_inflation` settings both trade a little accuracy on typical seasons for materially less-bad failure on an atypical one, and the bootstrap-based probabilistic targets (Section 6.6) give an explicit, checkable confidence read-out (e.g. how concentrated the peak-week distribution is) that a downstream user can inspect rather than trusting a point forecast blindly.

A further, more subtle calibration finding: PIT/coverage diagnostics showed the raw residual model is *overconfident*, never underconfident, on the hard season. This asymmetry is itself informative - it suggests the model's uncertainty estimate is anchored too tightly to the specific variance observed in its 2 training seasons, a generic risk whenever a quantile model is trained on a small number of exchangeable-in-theory-but-not-in-practice groups, and is the reason we recommend any future extension of this work (Section 10) prioritise either more training seasons or an explicit season-regime indicator over further tuning of the current 3-season-calibrated layers.

10. Conclusions

The CP-DWRE is an original, four-component hybrid forecasting framework - a covariate-forced renewal-equation core, zone-level hierarchical calibration, a LightGBM residual quantile correction, and a parametric-bootstrap probabilistic ensemble - built specifically for the MAEPiMS Challenge 2026's synthetic Nigeria influenza dataset. Every reported result is validated leave-one-season-out across all 3 possible splits, a discipline that directly caught and corrected an overclaimed single-split result during development (Section 6.2) and surfaced a genuine, non-trivial Pareto improvement in probabilistic calibration (Section 6.3). The model performs well - often very well - on seasons resembling its training data, and degrades in a specific, diagnosed, and partially mitigated way on the one season (2024/25) that does not. We consider that honest characterisation of the model's limits, evidenced across every one of its four components rather than asserted once, to be as much a contribution of this submission as the point-forecast accuracy numbers themselves.

Priority directions for future work: (1) a rolling-origin backtest that forecasts forward from partway through a season using only that season's own early weeks, to directly test the nowcasting-vs-forecasting distinction flagged in Section 8.2; (2) an explicit season-severity-regime indicator, learned or elicited, that the calibration and residual layers could condition on rather than implicitly averaging across; and (3) replacing the Monte Carlo independence assumption (Section 2) in regional/national aggregation with an explicitly correlated sampling scheme once enough seasons exist to estimate that correlation structure.

References

- Bracher, J., Ray, E. L., Gneiting, T., & Reich, N. G. (2021). Evaluating epidemic forecasts in an interval format. *PLOS Computational Biology*, 17(2), e1008618.
- Cori, A., Ferguson, N. M., Fraser, C., & Cauchemez, S. (2013). A new framework and software to estimate time-varying reproduction numbers during epidemics. *American Journal of Epidemiology*, 178(9), 1505-1512.
- Fraser, C. (2007). Estimating individual and household reproduction numbers in an emerging epidemic. *PLOS ONE*, 2(8), e758.
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359-378.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30.
- Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33-50.
- Seber, G. A. F., & Wild, C. J. (2003). *Nonlinear Regression*. Wiley.
- The Adejimi Adeniji Foundation (2026). *MAEPiMS Challenge 2026: Nigeria Influenza Forecasting Challenge* [Problem statement and synthetic dataset].